NEGARO MANZIR

Full-Stack Web Developer · Computer Science Student

Ambo / Arba Minch, Ethiopia | negaromanzir@gmail.com | (+251) 931 457 595
Portfolio: negaromanzir-lab.vercel.app | GitHub: github.com/negaromanzir-lab | Telegram: @negoman

PROFESSIONAL SUMMARY

Computer Science student at Arba Minch University and proactive Full-Stack Web Developer with practical expertise building responsive frontend interfaces, RESTful web services, and relational/document database architectures. Demonstrated experience implementing role-based access control, stateful APIs, vector retrieval systems (RAG), and dual web/desktop platforms using React, Node.js, Express, PostgreSQL, and MongoDB. Committed to clean code, robust system design, and continuous engineering growth.

TECHNICAL SKILLS

Languages: JavaScript (ES6+), TypeScript, Python, C++, Java, PHP, HTML5, CSS3, SQL
Frontend: React.js, React Router, Vite, Tailwind CSS, Responsive Design, Component Architecture
Backend & APIs: Node.js, Express.js, RESTful API Design, Server-Sent Events (SSE), JWT & Session Authentication
Databases & Tools: PostgreSQL, pgvector, MongoDB (Mongoose), MySQL, Git, GitHub, Bun, Electron, Linux/Bash
Security & Architecture: OWASP Top 10 Awareness, Input Validation, Role-Based Access Control (RBAC), Practical Auditing

KEY ENGINEERING PROJECTS

DxValley Incubation Center Management System (ICMS) **[Node.js, Express, PostgreSQL, EJS, Electron, Tailwind]**
Live: dxvalley-icms.onrender.com | Repo: github.com/negaromanzir-lab/Business-Incubator-Platform-DBMS
• Architected an operations platform connecting founder web applications with an Electron desktop manager interface over a shared PostgreSQL database.
• Implemented a 3-stage venture state machine (Idea -> MVP -> Scale-Up) with automated notification triggers for team members upon stage review.
• Designed normalized relational schemas with foreign key integrity governing entrepreneurs, mentors, facility bookings, and investor matching.

Document-Aware AI Assistant & ChatBot (RAG Architecture) **[React, Express, Node.js, PostgreSQL, pgvector]**
Repo: github.com/negaromanzir-lab/ChatBot-app
• Built a production-ready conversational AI platform featuring Retrieval-Augmented Generation (RAG) over user-uploaded documents (PDF, DOCX, text).
• Implemented server-side vector embedding and top-K cosine similarity retrieval via PostgreSQL pgvector, scoping results strictly per user conversation.
• Provided real-time Server-Sent Events (SSE) streaming responses with browser abort signals and per-user daily usage quota enforcement.

Real-Time Employee Management System (EMS) **[React Router 8, TypeScript, Express, MongoDB, Bun, Inngest]**
Repo: github.com/negaromanzir-lab/AI-Powered-Real-time-Employee-Management-System
• Engineered a modular workforce portal with 3-tier RBAC (Admin, Department Head, Employee) secured via httpOnly JWT cookies.
• Built clock-in/clock-out attendance logging, department leave policy approval pipelines, and task assignment tracking.
• Configured event-driven Inngest background workers to process automated operational insights without blocking the REST API.

EDUCATION & VERIFIED CERTIFICATIONS

Bachelor of Science in Computer Science · Arba Minch University, Ethiopia (2024 – 2028, In Progress)
• Core Coursework: Data Structures, Algorithms, Database Systems, Software Engineering, Computer Networks.

Udacity Nanodegree — Programming Fundamentals · Completed July 18, 2025
Verification: udacity.com/certificate/e/9ed3f802-4510-11f0-9ce2-efed14404efb

Udacity Nanodegree — Android Developer Fundamentals · Completed August 24, 2025
Verification: udacity.com/certificate/e/16f8e10a-63b4-11f0-b337-5f5de1371085

Udacity Nanodegree — Artificial Intelligence Fundamentals · Completed August 27, 2025
Verification: udacity.com/certificate/e/3ecd5892-7849-11f0-8ae5-d366902f426e

TECHNICAL OUTREACH & ACTIVITIES

Technical Content Creator & Educator · YouTube Channel (@negaromanzir) · 2023 – Present
• Produce technical video tutorials covering JavaScript, full-stack web architectures, API integration, and ethical security fundamentals.